

STATE-OF-THE-ART REVIEW

Screening Strategies for Identification of Transthyretin Amyloid Cardiomyopathy



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HIGHLIGHTS

- Multiple models have been developed to identify patients at high risk of ATTR-CM.
- Significant gaps remain in the evidence base for ATTR-CM screening strategies.
- This review synthesizes existing published reports and discusses screening approaches.
- Future work should better evaluate implementation and cost-effectiveness for screening strategies.

ABSTRACT

Transthyretin amyloid cardiomyopathy (ATTR-CM) remains underdiagnosed relative to population estimates in at-risk populations. Efforts have identified clinical features that identify risk for the diagnosis of ATTR-CM and have developed associated risk prediction models to identify risk of the presence of clinical ATTR-CM disease. Prospective validation and broad implementation of these data across health care systems have been limited. In this context, there is an unmet need to synthesize existing risk modeling published reports and discuss implementation approaches to identifying ATTR-CM. In this review, we summarize results from risk prediction efforts to date, identify remaining gaps in evidence, and describe potential opportunistic health system strategies to integrate systematic ATTR-CM risk prediction and screening. (JACC Heart Fail. 2026;14:103018) © 2026 by the American College of Cardiology Foundation.

Transthyretin amyloid cardiomyopathy (ATTR-CM) is an infiltrative cardiomyopathy and an increasingly recognized cause of heart failure (HF) in older adults. The condition can present heterogeneously with multisystem symptoms and signs including cardiac (eg, conduction disease, arrhythmias, syncope, HF) and extracardiac (eg, carpal tunnel, spinal stenosis, polyneuropathy, autonomic dysfunction) manifestations.¹ There are 2 forms of ATTR-CM: variant-associated (ATTRv) and wild-type (ATTRwt). Over the past 2 decades, there have been tremendous advances progress in the noninvasive diagnosis^{2,3} and targeted treatment^{4,5} of ATTR-CM with the U.S. Food and Drug

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**ABBREVIATIONS
AND ACRONYMS****AI** = artificial intelligence**AL** = light chain amyloidosis**ATTR-CM** = transthyretin
amyloid cardiomyopathy**AUC** = area under the curve**ECG** = electrocardiogram**EHR** = electronic health record**HF** = heart failure**HFpEF** = heart failure with
preserved ejection fraction**LVEF** = left ventricular
ejection fraction**LVH** = left ventricular
hypertrophy**PYP** = technetium-99m
pyrophosphate cardiac
scintigraphy**RWT** = relative wall thickness

Administration (FDA) approval of transthyretin stabilizers^{6,7} and gene silencers.⁸ Given the mechanisms by which these agents act, currently approved therapies carry the greatest benefit when used earlier in the disease course.⁶

Even with increasing recognition, ATTR-CM remains underdiagnosed on a population-level; for instance, targeted screening studies indicate prevalence rates of 6.3% to 13.3% in older patients with heart failure with preserved ejection fraction (HFpEF), but ATTR-CM continues to elude diagnosis in many of these patients, resulting in much lower detected prevalence rates in national databases.^{1,9,10} Moreover, diagnosis may be relatively delayed and late in the disease process.¹¹⁻¹⁴

Without accurate and timely diagnosis, ATTR-CM is associated with high morbidity and mortality. This has prompted numerous efforts to identify features that suggest high risk of ATTR-CM and to develop models to guide diagnostic testing for ATTR-CM. From these efforts, a wide range of predictive features has arisen from different derivation data sets with substantial variability in final model components. There remains uncertainty regarding the optimal models for ATTR-CM risk prediction and best practices for implementation of risk prediction models at scale.

This review aims to summarize risk prediction findings across studies, describe remaining evidence gaps, and outline potential health system strategies for systematic ATTR-CM risk prediction and screening. The format of the paper will be as follows: 1) review of established models/scores (echocardiographic metrics, then electronic health record-based models of clinical features, machine learning models [clinical, echocardiographic, and electrocardiographic], and other notable models); 2) comparison studies of models; 3) key gaps in evidence for screening models; and 4) potential health system screening strategies for implementation and testing.

MODEL AND SCORE DEVELOPMENT

ECHOCARDIOGRAPHIC PREDICTORS OF ATTR-CM. Early efforts to predict the likelihood of ATTR-CM focused primarily on echocardiographic features in predefined populations (ie, suspected ATTR-CM and/or left ventricular hypertrophy [LVH] >12 mm) with relatively high disease prevalence.¹⁵⁻¹⁸ Risk-enhancing features have included E/e', relative wall thickness (RWT), tricuspid annular plane systolic

excursion, and longitudinal strain (Table 1). Models incorporating these features, coupled at times with electrocardiographic (ECG) and/or demographic characteristics, exhibited receiver operating curve areas under the curve (AUC) of ~0.80-0.93 in derivation population cohorts.

In one of the earliest studies, Boldrini et al¹⁵ evaluated echocardiographic data for 1,187 patients with: 1) proven light chain (AL) amyloidosis (n = 494); or 2) suspected amyloid and left ventricular wall thickness ≥12 mm (n = 978). The analysis of this second cohort produced a multiparametric score including RWT, E/e', longitudinal strain, tricuspid annular plane systolic excursion, and septal apical-to-base; this was termed the "IWT score" for its source cohort of patients with increased wall thickness, including 339 patients with ATTR-CM. This study was multisite across 3 countries (United Kingdom, Italy, and Spain) and included a broad set of advanced echocardiographic features; it was limited by its focus on a highly selected referral population at specialized amyloid centers.

In one of the largest ATTR-CM-specific studies, Cuddy et al¹⁹ examined 598 patients from 3 centers referred for technetium-99m pyrophosphate cardiac scintigraphy (PYP) scan due to suspicion for ATTR-CM. In a training (n = 173) and validation (n = 425) approach, 3 features were found to be the strongest predictors of ATTR-CM: age, inferolateral wall thickness, and basal longitudinal strain; with a combination producing an AUC of 0.87 (95% CI: 0.83-0.90) in the validation cohort. This study was also strengthened by its deep echocardiographic phenotyping and multicenter design but was limited by its cohort of a selected referral population at major urban academic amyloidosis centers, by an exclusive focus on echocardiographic features, and by some data missingness.

Continued work over the subsequent years sought to further validate and also establish novel echocardiographic features that could signal an increased likelihood of ATTR-CM among those with HF and/or LVH (>12 mm) (Table 1). For instance, through this work, relative apical sparing on global longitudinal strain has been established as associated with a diagnosis of ATTR-CM in the appropriate clinical context,²⁰⁻²² but it should be recognized that its sensitivity and specificity with specific cutoff values may be limited, particularly in concomitant conditions such as aortic stenosis or chronic kidney disease.²³⁻²⁸ In aggregate, across all echocardiographic-focused studies, the 2 most common characteristics for enhanced ATTR-CM risk were

TABLE 1 Select Studies Producing Models of Risk Features Associated With Diagnosis of ATTR-CM

First Author, Year	Base Population Criteria	Population Cohort, n	ATTR-CM Diagnoses in Cohort, n	ATTR-CM Known Prevalence, %	Key Model/Score Features	Performance Metrics
Cariou et al, ¹⁸ 2017	LVH	114 (derivation) 34 (validation)	31 (+18 AL amyloidosis) 14 (validation)	27.2 (derivation) 41.2 (validation)	- Systolic arterial pressure <130 mm Hg - PR duration >200 ms - Sokolow index <12 mV - Diastolic LV posterior thickness >13 mm - E/Ea ratio >10 - Global longitudinal strain >-12% - Sum of basal longitudinal strain >-47%	AUC: 0.93 (95% CI: 0.89-0.98) Sensitivity: 90% Specificity: 81%
IWT score, Boldrini et al, ¹⁵ 2020	Proven AL amyloidosis or those with LV wall thickness ≥12 mm with amyloid suspected	1,187	339 (+332 AL amyloidosis)	28.6	- RWT - E/e' - Longitudinal strain - Tricuspid annular plane systolic excursion - Septal apical-to-base	AUC: 0.87 (95% CI: 0.85-0.90) Sensitivity: 82% Specificity: 72%
AMyli score, Aimo et al, ¹⁶ 2020	LV wall thickness ≥12 mm	184 (derivation) 691 (validation)	79 (derivation) (+ separate AL amyloidosis cohort)	42.9	- RWT - E/e'	AUC: 0.75 (Optimal sensitivity/specificity: N/P)
Zhang et al, ¹⁷ 2021	Patients who underwent cardiac biopsy for suspected amyloidosis	242	60 (+53 AL amyloidosis)	24.8	- Age - RWT - E/e' - Low limb lead voltage	AUC: 0.88 Sensitivity: 79% Specificity: 84%
Mayo ATTR-CM score (TCAS), 2022 ²⁹	Derivation: HFpEF and suspected ATTR-CM referred for PYP Validation: older patients with HFpEF and LVH ≥12 mm and separate HFpEF cohort referred for PYP	416 (derivation) 250 + 286 + 66 (validation)	189 (derivation) 120 + 17 + 26 (validation)	45.4 (derivation) 27.1 (validation)	- Age ≥80 y (or 70-79 y; or 60-69 y) - Male sex - Absence of HTN diagnosis - LVEF <60% - Posterior wall thickness ≥1.2 cm - RWT >0.57	AUC: 0.89 (95% CI: 0.86-0.92) Sensitivity: 93% Specificity: 62%
Cuddy et al, ¹⁹ 2022	Patients referred for PYP to detect ATTR-CM	598 (173 training + 425 validation)	324	54.2	- Age - Inferolateral wall thickness - Average basal longitudinal strain	AUC: 0.90 (validation) AUC: 0.87; 95% CI: 0.83-0.90)
Amberg et al, ⁷³ 2022	IVSd ≥14 mm (ATTR-CM cohort and LVH cohort)	167 (derivation) 20 (validation)	102 (derivation) 20 (validation)	61.1 (derivation) 100 (validation)	RWT/S-wave from aVR	AUC: 0.95 Sensitivity: 97% Specificity: 90%
T-Amylo, 2023 ⁴⁸	IVSd ≥1.2cm + ≥1 red flag (consecutive patients who underwent PYP for suspected ATTR-CM)	227 (derivation) 895 (validation)	108 (derivation) 396 (validation)	47.6 (derivation) 44.2 (validation)	- Carpal tunnel syndrome - Male sex - LVH by IVSd ≥1.6 cm - Low QRS voltage^a - Age ≥80 y	AUC: 0.92 (95% CI: 0.88-0.95; model) AUC: 0.86 (95% CI: 0.81-0.90; simplified score) Sensitivity: 88% Specificity: 63%
Moya et al, ¹³ 2023	HF diagnosis (by ICD-10, internal labeling, and/or NLP detection)	3,127	103	3.3	- AF - Heart block - Joint disorders - OA - Soft tissue disorders	NA
Suh et al, ⁷⁴ 2023	Confirmed ATTR-CM or HF (by ICD-10 codes)	7,651	31	0.41	Propensity model with 16 inputs: Black race, diuretic use, elevated BNP, nonvalvular AF/flutter, cardiomegaly, carpal tunnel syndrome, CKD, heart block, HFpEF, joint disorders, lumbar spinal stenosis, neuropathy, OA, pericardial effusion, refraction and accommodation disorder, and synovitis/tenosynovitis	AUC: 0.88 Sensitivity: 72% Specificity: 95%

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significant LVH (>12-16 mm) and RWT (>0.6). Performance of all features and associated models were impacted by the relatively high ATTR-CM disease prevalence of derivation cohorts.

MAYO ATTR-CM SCORE OR TRANSTHYRETIN CARDIAC AMYLOIDOSIS SCORE. Beyond echocardiographic features alone, subsequent studies began to investigate combination models to predict risk of

TABLE 1 Continued

First Author, Year	Base Population Criteria	Population Cohort, n	ATTR-CM Diagnoses in Cohort, n	ATTR-CM Known Prevalence, %	Key Model/Score Features	Performance Metrics
Nagai et al, ⁷⁵ 2023	Older patients (aged ≥ 65 y) who underwent PYP to evaluate for suspected ATTR-CM	89	34	38.2	- Posterior wall thickness - RVLS - LV longitudinal strain apical/basal ratio	AUC: 0.90 (with first 2 metrics alone) Score using 3 metrics: Sensitivity: 85% Specificity: 93%
EstimATTR, 2024 ³¹	ICD codes for ATTRwt-CM and HF Age >50 y Matched on age and sex	672 (derivation) 538 + 16,118 + 4,221 (validation cohorts)	317 (derivation) 280 + 8,274 + 2,187 (validation cohorts)	47.2 (derivation) 51.5 (validation)	- Carpal tunnel syndrome - Pericardial effusion - HFpEF - LVH - Atraumatic tendon rupture - Elevated enzymes - Conduction disease - Arrhythmias - Polyneuropathy - Lumbar spinal stenosis - Degenerative joint disorders	AUC: 0.82 Sensitivity: 77% Specificity: 72%
AMY score, 2024 ⁷⁶	ATTRwt-CM diagnoses + HFpEF cohort Patients undergoing work-up for ATTRwt-CM	82 (derivation) 561 (validation)	41 (derivation) 142 (validation)	50 (derivation) 25.3 (validation)	- IVSd >14 mm - E/e' >14 - Absence of obesity	AUC: 0.91 (95% CI: 0.84-0.97) Sensitivity: 78% Specificity: 90%
Shingu et al, ⁷⁷ 2025	Suspected ATTR amyloidosis undergoing PYP	135	62	45.9	- Troponin I/T ≥ 0.06 ng/mL - Absence of HTN - Existence of peripheral entrapment neuropathy (carpal tunnel syndrome or spinal canal stenosis) - Conduction disturbance^a - LVH	AUC: 0.82 (95% CI: 0.75-0.89) Sensitivity: 63% Specificity: 84%

^aThe presence of QRS complex ≥ 120 ms, first-degree atrioventricular block, higher degree of atrioventricular block, or presence of pacemaker implantation.

AF = atrial fibrillation; AL = light chain amyloidosis; AMY = Absence of obesity, Myocardial hypertrophy, Diastolic dysfunction; AMYLI = amyloidosis index; ATTR-CM = transthyretin amyloid cardiomyopathy; ATTRwt-CM = transthyretin amyloid wild-type cardiomyopathy; AUC = area under the curve; BNP = B-type natriuretic peptide; CKD = chronic kidney disease; HF = heart failure; HFpEF = heart failure with preserved ejection fraction; HTN = hypertension; ICD = International Classification of Diseases; IVSd = interventricular septum thickness; IWT = increased wall thickness; LV = left ventricular; LVEF = left ventricular ejection fraction; LVH = left ventricular hypertrophy; OA = osteoarthritis; PYP = technetium-99m pyrophosphate cardiac scintigraphy; RVLS = right ventricular longitudinal strain; RWT = relative wall thickness; TCAS = transthyretin cardiac amyloidosis score.

ATTR-CM diagnosis. One of the first and largest efforts, drawing from referral cohorts and population-level studies of ATTR-CM screening, was conducted through the Mayo Clinic.^{10,29} A PYP-referral cohort (n = 666) (2013-2020) was developed in southeastern Minnesota, USA (Mayo Clinic and surrounding clinics) that included those with confirmed HFpEF and a PYP scan within 1 year. The validation cohorts included an external PYP-referral cohort and a community-based HFpEF cohort of older patients (≥ 60 years of age) with HFpEF and LVH (≥ 12 mm) who underwent a systematic screening strategy (n = 286) (ATTR-CM was diagnosed in 6.3%).

Through the internal PYP-referral cohort (n = 666) and these validation cohorts, an ATTR-CM risk score was developed consisting of age, sex, absence of HTN diagnosis, left ventricular ejection fraction (LVEF) $<60\%$, posterior wall thickness ≥ 12 mm, and RWT >0.57 . The performance metrics were strong (derivation AUC: 0.89 [95% CI: 0.86-0.92]; $P < 0.001$; and derivation calibration: Hosmer-Lemeshow chi-square = 4.6; $P = 0.46$) and maintained in all

validation cohorts: positive predictive value of 25% and negative predictive value of 97% at score threshold ≥ 6 vs <6 at an ATTR-CM prevalence of 10%. The strongest risk-enhancing components of this score were advancing age (≥ 80 years old, 70-79 years old, 60-69 years old), male sex, and RWT >0.57 (Table 2).

Similar to the study by Cuddy et al,¹⁹ the Mayo clinic study was also limited to a selected referral population with clinical concern for ATTR-CM, but it did include a community-based validation cohort. The derivation and initial validation cohorts were mostly White participants ($\sim 94\%$), but there was an additional external validation in a more diverse population (37% Black). The study included both clinical and echocardiographic data but was limited by some echocardiographic missing data and ECG data missingness.

SIMPLIFIED, MACHINE LEARNING-ENABLED MODELING IN ELECTRONIC HEALTH RECORDS. Machine learning models have also been pursued to evaluate complex interactions between multiple input variables as a

TABLE 2 Prediction Metrics of Risk Features in Select Multivariable Risk Models for Diagnosis of ATTR-CM

	ORs in Final Models		
	Mayo (TCAS)	T-Amylo ^a	EstimATTR
	Suspected ATTR-CM (HFpEF, Referred for PYP)	Suspected ATTR-CM (IVSd ≥ 12 mm, ≥ 1 Red Flag, Referred for PYP)	Age >50 y and ICD Codes for ATTRwt-CM and/or HF
ATTR-CM prevalence			
Derivation	45.4%	47.6%	47.2%
Validation	27.1%	44.2%	51.5%
Age, y			
60-69	7.8 (2.2-27.4)	—	—
70-79	15.8 (4.8-51.6)	—	—
≥ 80	53.3 (15.2-186.6)	3.55	—
Male	7.2 (3.1-16.8)	8.85 (NA)	—
Carpal tunnel syndrome	—	8.40 (NA)	6.2 (5.0-7.7)
LVH			
≥ 12 mm	3.9 (1.8-8.4) ^b	—	3.0 (2.6-3.4)
≥ 16 mm	—	5.79 (NA)	—
RWT >0.57	4.6 (2.5-8.6)	—	—
Absence of HTN	3.33 (1.67-5)	—	—
LVEF $<60\%$	2.4 (1.4-4.2)	—	—
HFpEF	—	—	4.0 (3.4-4.6)
Pericardial effusion	—	—	5.8 (4.3-8.1)
Conduction diseases	—	—	2.6 (2.3-3.1)
Arrhythmias	—	—	2.4 (2.1-2.8)
Low QRS voltage	—	5.20 (NA)	—
Elevated serum enzymes	—	—	2.7 (2.3-3.3)
Atraumatic tendon rupture	—	—	2.9 (2.1-4.1)
Polyneuropathy	—	—	2.3 (1.9-2.8)
Lumbar spinal stenosis	—	—	1.8 (1.5-2.1)
Degenerative joint disorders	—	—	1.7 (1.5-2.0)

^aORs from simplified scoring model. ^bOR associated with posterior wall thickness ≥ 12 mm.
NA = not applicable; other abbreviations as in Table 1.

strategy to advance ATTR-CM risk prediction and identify new risk modifiers/combinations of modifiers. One of the largest efforts to date was the study by Huda et al³⁰ that used national medical claims and electronic health record (EHR) data sets combined with the Northwestern single-center data warehouse. Derivation was completed based on 1,071 cases and 1,071 HF controls and was validated in 4 additional cohorts including 9,673 cases and 48,805 controls. The predictive indexes varied (ie, AUC: 0.76-0.95) based on use of medical claims data alone or full EHR extracted data and based on prediction of ATTRwt-CM or cardiac amyloidosis (wt or v; attempting to exclude AL amyloidosis).

From this machine learning model, the group identified 11 predictive features to produce a simplified model (EstimATTR) with an AUC of 0.82 (and accuracy: 74%; sensitivity: 77%; specificity: 72%; positive predictive value: 71%; negative predictive value: 78%; false positive rate: 28%; false negative rate: 23%) in the derivation cohort and slightly lower

performance in validation cohorts.³¹ The 11 features were HFpEF, elevated serum enzymes (ie, troponin or N-terminal pro-B-type natriuretic peptide), conduction diseases, arrhythmias, pericardial effusion, LVH, carpal tunnel, degenerative joint disorders, lumbar spinal stenosis, atraumatic tendon rupture, and polyneuropathy.

This study was strengthened by its cohort size and extensive validation but was limited by a reliance on administrative data sets (ie, International Classification of Diseases [ICD] coding) and the assumptions of Bayesian statistical modeling.

MACHINE LEARNING MODELS WITH ECHOCARDIOGRAPHY AND ELECTROCARDIOGRAPHY. Machine learning models, including deep learning models, have also been developed exclusively using echocardiogram or ECG waveform/images (Table 3) for prediction of cardiac amyloidosis (CA) diagnosis (often including ATTR-CM and AL-cardiac amyloidosis). One of the first models, from Zhang et al,³² used 14,035 echocardiograms from a single center (University of

TABLE 3 Select AI Models Utilizing Electrocardiography and/or Echocardiography for the Diagnosis of Cardiac Amyloidosis

First Author, Year	Base Population	Population Cohort, n	CA Diagnoses in Cohort, n ^a	ECG/Echo Input	Model Architecture	Performance Metrics
Tison et al, ³³ 2019	Single-center ECG database	Total study: 36,186 ECGs (816 for amyloid analysis)	180 ECGs (~92 patients)	ECG	CNN and hidden Markov models	AUC: 0.86 (95% CI: 0.82-0.89)
Goto et al, ³⁵ 2021	Cases and matched controls with ECG from 3 centers	7,937 (training) 1,272 (validation)	480 (training) 117 (validation)	ECG	2D-CNN-based model	AUC: 0.85-0.91 Sensitivity: 52%-71%
	Cases and matched controls with echo from 5 centers	2,828 (training) 1,084 (validation)	410 (training) 199 (validation)	Echo	3D-CNN-based model	AUC: 0.89-1.00 Sensitivity: 67%
Grogan et al, ³⁶ 2021	Single-center CA cohort and matched controls (Separate validation study ³⁷)	4,995 total: 2,997 (training) 999 (test) 999 (validation)	530 ATTR-CM (and 1,839 AL): 1,525 (training) 508 (test) 508 (validation)	ECG	CNN-based model used in prior studies ⁷⁸	AUC: 0.91 (95% CI: 0.90-0.93) Sensitivity: 84% Specificity: 85%
EchoNet-LVH, Duffy et al, ^{34,79} 2022	Multiple echo databases (including ATTR-CM, HCM, and controls)	6,461 (training) 814 (validation) 3,160 (test)	950 (training) 116 (validation) 478 (test)	Echo	ResNet3D (a form of CNN modeling)	AUC: 0.83
LVH-net, Haimovich et al, ³⁸ 2023	Adults with LVH and/or known CA or HCM at a multisite academic health care system	34,258 (derivation) 16,451 (validation)	147 (derivation) 157 (validation)	ECG	CNN-based model	AUC: 0.95 (95% CI: 0.93-0.97) Sensitivity: 90% Specificity: 89%
Chang et al, ⁴⁰ 2024	Single-center echo database: CA cases and controls, high CA suspicion but ruled out	636 total: 486 (training) 150 (test)	231 total: 55 AL amyloidosis, 91 ATTR-CM, and 85 unspecified	Echo	Random forest model	AUC: 0.84 Sensitivity: 82% Specificity: 73%
EchoGo Amyloidosis, Slivnick et al, ^{39,41} 2025	Multisite echo database (training) Global, multisite registry (validation)	2,612 (training) 2,719 (validation)	1,358 (training) 597 (validation)	Echo	3D-CNN-based model	AUC: 0.93-0.98 Sensitivity: 85%-91% Specificity: 93%-96%
Oikonomou et al, ⁸⁰ 2025	Multisite echo and POCUS databases	8,460 (training/validation) 38,751 (test)	~3% of images (training/validation) ~0.2% of images (test)	Echo	3D-CNN-based model	AUC: 0.91-0.97 Sensitivity: 73%-100% Specificity: 77%-87%
ATTRACTnet, Jain et al, ⁴³ 2025	Multisite clinical, ECG, and echo databases	799 (internal test) 422 (external test)	165 (internal test) 96 (external test)	ECG	ResNet + tabular data including echo, demographics, etc	AUC: 0.77-0.85 Sensitivity: 56%-72% Specificity: 78%-92%

^aCA includes both AL amyloidosis and ATTR-CM.

2D = 2-dimensional; 3D = 3-dimensional; AI = artificial intelligence; CA = cardiac amyloidosis; CNN = convolutional neural networks; ECG = electrocardiogram; echo = echocardiogram; HCM = hypertrophic cardiomyopathy; POCUS = point-of-care ultrasonography; other abbreviations as in Table 1.

California-San Francisco) and trained convolutional neural network models for discernment of cardiac structure and disease, including cardiac amyloid (ie, AUC: 0.87).³²

A subsequent study by Tison et al³³ from the same center evaluated 36,186 ECGs and applied a combination of convolutional neural network and hidden Markov models to train machine learning models for cardiac structure and disease, including cardiac amyloid (ie, AUC: 0.86) with ~92 patients with cardiac amyloid diagnosis compared with controls.

Duffy et al³⁴ presented another model, EchoNet-LVH, that used thousands of echocardiography images (primarily parasternal long-axis or apical 4-chamber videos) from a total of 23,745 patients across multiple centers, including ~1,550 patients with diagnosed ATTR-CM. The deep learning

algorithm (EchoNet-LVH) was accurate in identifying LV wall thickness and demonstrated AUCs of 0.79-0.83 for classifying cases as consistent with a diagnosis of ATTR-CM. Several other studies have applied machine learning models to ECG^{33,35-38} or echocardiography images^{35,39-41} or the combination of ECG-machine learning modeling and simple tabular data,^{42,43} and they have demonstrated strong discrimination of ATTR-CM diagnosis compared with controls.

As of October 2025, there have been 3 FDA-cleared platforms as Software as a Medical Device (SaMD) for the detection of cardiac amyloidosis using artificial intelligence (AI)-echocardiography: EchoGo Amyloidosis (Ultromics) (AUC: 0.86%-0.98; sensitivity 85%-91%, specificity 93%-96%),^{39,41} EchoNet-LVH (InVision) (AUC: 0.79-0.83),^{34,44} and Us2.ca

(Us2ai) (AUC: 0.87-0.98; conference presentation data prepublication).⁴⁵⁻⁴⁷

ADDITIONAL PREDICTION SCORES. Several other investigations have generated prediction scores with a combination of clinical and echocardiographic features (Table 1). One retrospective study from a cohort of patients at a center in Belgium found the strongest cardiac predictor of ATTR-CM diagnosis was atrial fibrillation (AF), the strongest noncardiac predictor was carpal tunnel syndrome, and the strongest combination of predictors was AF, joint disorders, and HFpEF.¹³

The T-Amylo score, developed through a cohort of centers in Spain, is another notable study given its size and methodology.⁴⁸ The study focused on consecutive patients who underwent PYP scan for suspected ATTR-CM (based on interventricular septum thickness ≥ 1.2 cm + ≥ 1 red flag sign/symptom). The analysis used 227 derivation patients (48% ATTR-CM) from 2 centers and 895 validation patients (44% ATTR-CM) from multiple centers to produce a model with an AUC of 0.92 and simplified score of 0.86. The score consisted of age, sex, carpal tunnel syndrome, interventricular septum in diastole thickness of ≥ 1.6 cm, and low QRS interval voltage; of these, male sex and carpal tunnel had the highest influence on risk prediction. The simplified score (and the statistical model that it was based on) had similarly excellent performance across validation cohort and specific clinical subpopulations including hypertensive heart disease, severe aortic stenosis, and HFpEF, but notable limitations were a paucity of variant patients in the derivation cohort and likely less sensitivity in females, who have typically lower wall thickness than males.

COMPARISON STUDIES

There have been very few studies that have directly compared prediction models in distinct cohorts. One such study compared the Mayo ATTR-CM score and T-Amylo score in a cohort of 476 patients who underwent PYP scan for suspected ATTR-CM.⁴⁹ The cohort was notably diverse compared with prior derivation/validation study cohorts: 65% non-Hispanic Black, 20% Hispanic, and 45% female sex. The study found the Mayo ATTR-CM score to have superior predictive performance (AUC: 0.86; 95% CI: 0.83-0.90) compared with the T-Amylo score (AUC: 0.75; 95% CI: 0.71-0.80). Further, the study confirmed several features to be independent predictors of positive ATTR-CM testing in its own analysis, including age ≥ 74 years, male sex, history of carpal tunnel syndrome, LVEF $< 55\%$, posterior wall

thickness > 12 mm, and RWT > 0.57 . Some notable comorbidities suggested negative PYP testing for ATTR-CM including hemoglobin < 10 mg/dL, glomerular filtration rate < 30 mL/min/1.73 m², and coronary artery disease.

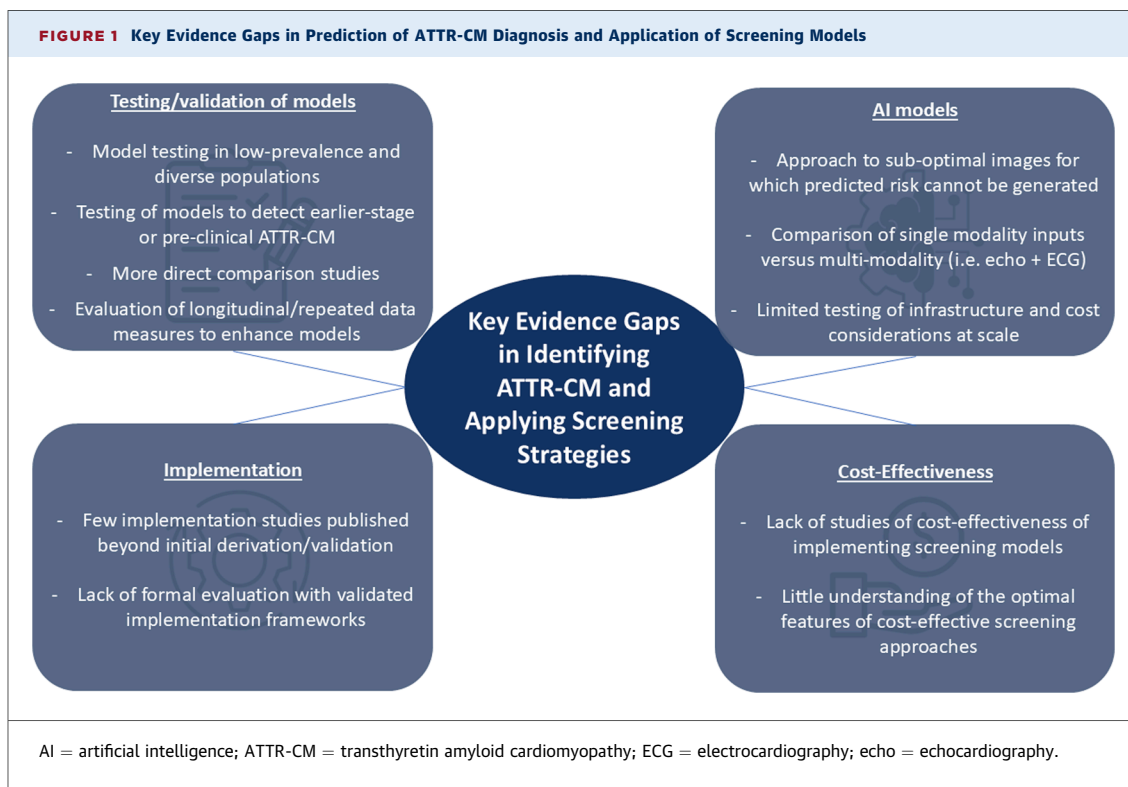
More recently, Hourmozdi et al⁵⁰ offered another direct comparison study that evaluated 4 algorithms (Mayo ATTR-CM,²⁹ Huda et al³⁰ model, and 2 deep learning echocardiogram-based models^{34,41}) in 176 ATTR-CM cases and 3,192 control patients. In the primary analysis of interpretable images, the 2 deep learning echocardiogram-based models (AUC: 0.88 and 0.92) outperformed the Mayo ATTR-CM model (0.79) and the Huda et al³⁰ (AUC: 0.49) in a broad population of matched controls of patients with a diagnosis code for HF. Although the cohort was predominantly self-identified as White (79.2%), bias-auditing analyses indicated the models met fairness criteria for equal opportunity among Black patients.

Of note, the deep learning AI image-based model could not generate predictions in 15% to 20% of patients due to factors such as “insufficient clip length, low image quality, insufficient videos by view classification, low predictive confidence, or uncertainty,” and therefore those patients (1,432 overall including 42 cases) were excluded from the primary analysis; sensitivity analyses with the full cohort (with the missing predictions imputed as “0,” negative) demonstrated similar AUCs across the 2 deep learning models and the Mayo ATTR-CM model.

Although the deep learning models with echocardiography had better model performance than the Mayo score in the primary analysis, the population studied in the validation differed from the study population under which the Mayo score was developed and validated, which likely explains the lower performance of the Mayo score in this data set compared with other external validation studies.

KEY GAPS IN EVIDENCE SUPPORTING SCREENING PROTOCOLS

Despite numerous studies focused on ATTR-CM risk models, there remain significant gaps in the field (Figure 1).⁵¹ First, most studies have been limited to relatively high prevalence cohorts; although several analyses have included validation cohorts, broad representativeness of the cohorts and consistent subgroup performance remain limited or only partially achieved.^{52,53} There is accordingly limited generalizability, so future studies should evaluate performance metrics across a range of subpopulations (ie, broad populations such as men > 60 years of age; populations with several risk features;



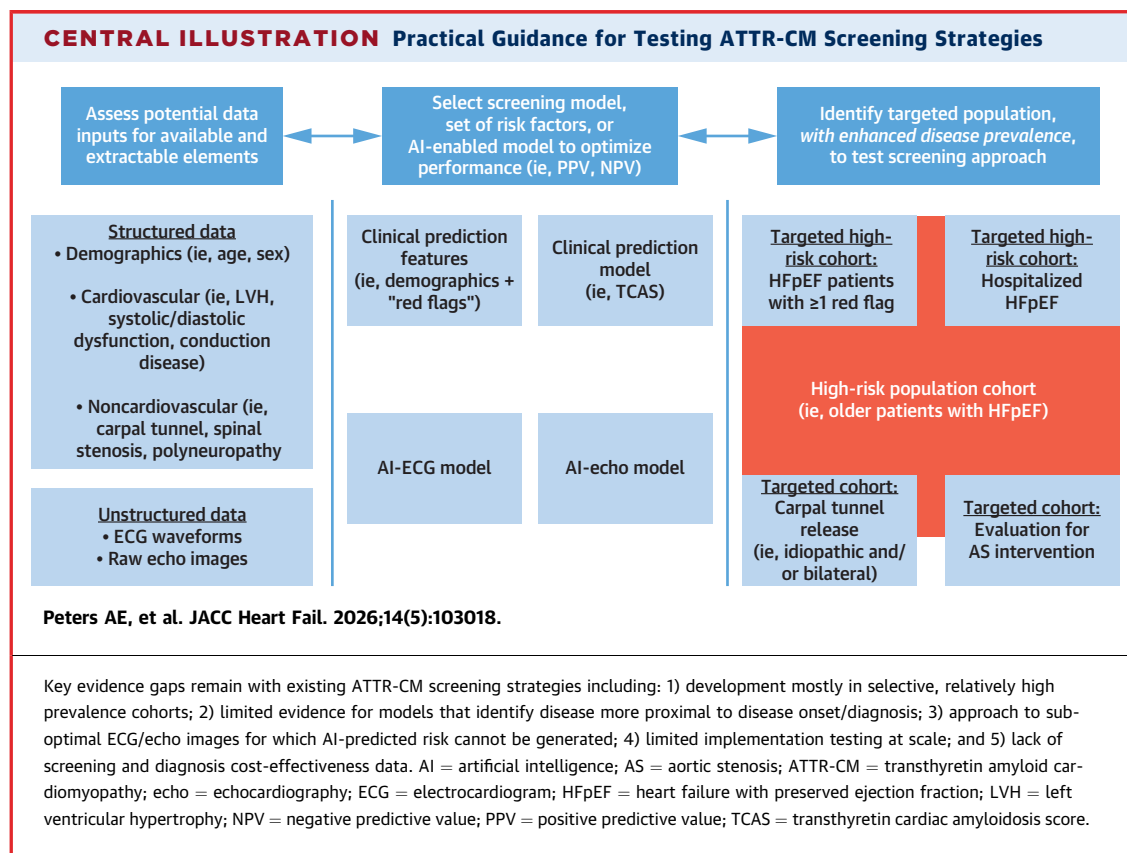
and highly selected, high-risk populations with numerous risk factors). The existing models have not been well-studied specifically in cohorts with low disease prevalence (where false positive rates may be higher) or in several ATTR-CM subpopulations such as patients with reduced LVEF (ie, HFrEF), which represents an important phenotype particularly in Black patients.⁵⁴

Further, accurate and replicable (derivation/validation) estimates for sensitivity, specificity, positive predictive value, and negative predictive value are still lacking for some of the risk features and risk models for ATTR-CM. Several of the reviewed studies either included limited metrics to evaluate the modeling (ie, AUC alone) or did not effectively validate the model findings. These additional data would support selection of implementation strategies and cost-effectiveness analyses.

Second, the models to date have focused on features associated with existing diagnosis or de novo diagnosis in the setting of high clinical concern. Few studies have systematically assessed for features that may be present more proximal to disease onset/diagnosis—ie, predictors of the future onset/

diagnosis of ATTR-CM (ie, subclinical or undiagnosed clinical, symptomatic ATTR-CM) as opposed to coinciding with diagnosis. Targeted studies have shown that the musculoskeletal manifestations of ATTR (such as, carpal tunnel syndrome) are known to precede overt ATTR-CM by 5 to 10 years,⁵⁵ but more work is required to determine other associated risk features that can reliably and consistently predict future development of symptomatic, clinical ATTR-CM. Serial, temporal assessments of AI ECG/echocardiography models are starting to be investigated for this purpose with some promise.⁵⁶ This is particularly important given the recent advances in targeted treatments,^{4,5} which still act by slowing down disease progression (as opposed to reversing disease damage), with evidence that the greatest benefits occur with early treatment.⁶

Third, cost-effectiveness data on applying risk modeling to populations for guiding definitive testing and treatment—a major priority for health systems—has been limited. Understanding cost-effectiveness is a critical aspect of developing and implementing broad population-based screening strategies for disease for multiple reasons (eg, limited



resources and opportunity cost).^{57,58} This is especially important for ATTR-CM screening strategies, given the wide range of potential screening populations (eg, HFpEF, aortic stenosis, carpal tunnel syndrome, etc), which represent a significant proportion of older adults and thus would take substantial effort and financial means to screen these populations.

One of the few studies on this topic suggested that “universal systemic screening” with PYP scans performed in an at-risk population (≥ 60 years of age, HFpEF, LVH ≥ 12 mm) might not generate a high incremental cost-effectiveness ratio considering the tafamidis costs and quality adjusted life-years achieved.⁵⁹ However, this assertion was based on multiple assumptions and derivation data sources, the inherent limitations of Markov modeling for cost-effectiveness, a single point estimate for ATTR-CM prevalence (6.8%), and inclusion of cost/benefit for only 1 branded drug (tafamidis). Other studies have focused more on comparing the cost-effectiveness of particular imaging strategies (PYP, cardiac magnetic resonance, PYP + cardiac magnetic resonance, endomyocardial biopsy) and have found PYP-only strategies may be most cost-effective, but they

relied on a hypothetical cost scenario for tafamidis (\$16,000 annually; alternative supplemental analyses with real-world branded pricing suggested low cost-effectiveness).⁶⁰

In this setting, much more work is necessary to better understand the optimal features of a cost-effective population-based screening approach (risk criteria, target population, screening strategy, deployment of treatment and resources), which may very well differ depending on the health system and the population it serves. The eventual transition to generic therapies will then alter the implementation cost-effectiveness paradigm.

POTENTIAL HEALTH SYSTEM ATTR-CM SCREENING STRATEGIES

Practical implementation of the described risk prediction findings and future work could be enacted in multiple ways (Central Illustration). Some considerations for selecting a screening strategy in ATTR-CM include scalability, compatibility with the EHR infrastructure of a health system, utilization of findings from multiple studies/risk scores, limited risk of inappropriate patient exclusions due to data

availability/interpretability, and costs/resources necessary for implementation. The effectiveness of any individual approach or combination of strategies will likely depend on the characteristics of each health system, such as its available resources.

SCREENING BY ESTABLISHED CLINICAL RISK FEATURES.

One screening strategy was developed by an expert panel in 2019, proposing PYP screening should be applied to men >65 years of age or women >70 years of age with HF or a “red flag” sign/symptom and LVH ≥ 14 mm.²¹ The red flags were defined as reduced longitudinal strain with apical sparing; echocardiographic LVH without ECG LVH; atrioventricular block and LVH; echocardiographic hypertrophic phenotype; cardiac magnetic resonance infiltrative cardiomyopathy signs; polyneuropathy and/or dysautonomia; bilateral carpal tunnel syndrome; and multiple, mild troponin elevations. This strategy would capture many of the known risk features for ATTR-CM and can effectively guide clinical practice, but it has not yet been systemically tested in practice, does not include several risk features (ie, pericardial effusion, RWT, absence of hypertension), and does not distinguish degree of individual patient risk (ie, very high, high, and moderate risk).

Of note, the recently completed SCAN-MP (Screening for Cardiac Amyloidosis With Nuclear Imaging in Minority Populations) prospective screening study uses relatively simple modifications of the previous enrichment criteria but without additional red flag symptoms beyond HF and LVH.^{61,62} This study focused on self-identified Black or Caribbean Hispanic individuals ≥ 60 years of age with HF, LVH ≥ 12 mm, and LVEF >30%, excluding those with HF known primarily due to ischemic heart disease and/or valvular heart disease. They found an ATTR-cardiac amyloidosis prevalence of 6.66% among 646 participants (subset of 1,564 eligible participants who elected to participate). Genetic screening was also performed and found a prevalence of V142I allele of 5.6%, with over half of these patients (52.8%) also having clinical ATTRv-CM. These and other data support the consideration of population screening approaches that focus on genetic testing (with simultaneous or follow-up imaging/biopsy screening as needed).⁶³ Some initial efforts to this effect have been executed without a high positive screening rate (ie, 0% positive screening among targeted study cohort of 256 from a population of 10,233).⁶⁴

SCREENING WITH VALIDATED RISK MODELS.

Alternatively, health care systems could select a single validated risk model (with calculable risk

score) to better stratify risk of ATTR-CM diagnosis in broad populations (ie, all patients with dyspnea or all patients with confirmed HF) and pursue screening at 1 or multiple risk cutoffs. For instance, one could apply the Mayo risk score to the health care system’s HFpEF population (with or without additional risk features or suspected ATTR-CM, as in the derivation study) and estimate individual-level risk of diagnosis. Screening could then be applied at certain risk cutoff (ie, ≥ 6 points, correlating with $\geq 25\%$ positive predictive value) or multiple strata (ie, electronic patient and primary provider notification of potential risk for ≥ 5 points; and electronic notification and phone call for ≥ 7 points). This would be based in one of the most well-validated risk scores with relatively simple criteria and could be theoretically titrated to a particular cost-effectiveness or total resources of a health care system. This would, however, rely on the accuracy of existing billing diagnoses (ie, HFpEF), does not account for several risk features (ie, carpal tunnel syndrome, lumbar spinal stenosis, pericardial effusion, or low QRS voltage), and has only been shown to be effective in selected at-risk populations and not as a broader screening tool.

TARGETED SCREENING IN SPECIFIC, HIGH-RISK POPULATIONS.

Another screening option would be to target screening strategies in *more specific high-risk populations*, particularly those with established, high contact with medical care (such as patients hospitalized for HFpEF, those undergoing carpal tunnel release, and those being evaluated for aortic stenosis intervention). One of the most well-established of these approaches is systematic biopsy screening in high-risk carpal tunnel disease populations; notably, this has been applied prospectively in a single-center study in men ≥ 50 years of age and women ≥ 60 years of age undergoing carpal tunnel release surgery and proven to detect amyloid deposits in 10% of patients, with 20% of this subset demonstrating active cardiac involvement.⁶⁵

A further stratified approach with additional risk features (ie, bilateral symptoms, prior release surgery, spinal stenosis) has been proposed to guide systematic screening,⁶⁶ and genetic screening approaches in carpal tunnel populations have been described.⁶⁷ Since these seminal studies, iterations of this carpal tunnel biopsy screening approach have been employed at multiple sites.⁶⁸ Prospective screening studies have also been successfully performed in patients with aortic stenosis referred for transcatheter aortic valve replacement, identifying amyloid cardiomyopathy in 10% to 11% of patients.^{69,70} These population-targeted screening

strategies are evidence-based with clear protocols and expected results, and they can be conducted without significant bioinformatics capabilities and resources. Still, they are limited to specific risk referral populations alone, and they do not inherently adjust for other risk features.

SCREENING THROUGH COMBINATIONS OF ESTABLISHED METRICS AND MODELS. Traditional rules-based approaches or artificial intelligence techniques (ie, natural language processing for signs and symptom extraction from clinical notes, or machine learning prediction models) could also be applied to health system data sets, guided by multiple studies from the existing evidence base for risk quantification of ATTR-CM.

For instance, with the traditional rules-based approach, one could apply serial risk criteria (ie, age → history of HF → etc) to a large population to increase the estimated risk of ATTR-CM at each serial stage and the potential yield of screening. Diagnostic testing could then follow sequentially from very high to high to intermediate risk with stratified intensity of patient outreach and screening according to risk. This strategy would include many of the optimal features identified earlier for a screening approach because it would be scalable and adaptable to any health care system (ie, flexible to exclude features not accessible in a particular EHR).

Further, it could leverage findings from multiple studies (ie, combine predictors from Mayo, T-Amylo, and/or AI-echocardiography) and would be relatively resilient to missing data/coding limitations and to residual risk in a patient without a particular risk feature, given there is no complete exclusion of a patient for lacking a diagnosis (ie, a patient with missing diagnosis is shifted to an alternate arm of the algorithm and further screened for risk features).

Additionally, it could be applied to a broad adult population, and it could be titrated for sensitivity/specificity and associated resource utilization. It could be further supported by natural language processing to extract signs, symptoms, and diagnoses to augment the utilization of existing EHR data. Key limitations to this strategy include that this approach has not yet been evaluated and validated, and combining risk factors from different models may not demonstrate similar performance when collated.

SCREENING THROUGH AI-AUGMENTED CLINICAL, ECG, AND/OR ECHOCARDIOGRAPHIC DATA. Alternatively, AI-dominant models based on ECG or echocardiography interpretation (Table 3) could be scaled across a health system population, using images obtained in normal clinical workflow or systematic image

screening approaches. Approaches with select models are being actively implemented (and tested) at multiple institutions.³⁹ These AI risk prediction models could be used in isolation or paired with data regarding known risk factors from studies described in this review; however, notably, they require logistical and technical support for scaling across health systems.

The recently conducted Cardiac Amyloidosis Discovery Trial was one of the first studies to prospectively study application of an AI-augmented detection model (ATTRACTnet) (described earlier and in Table 3) to recruit and prospectively screen patients for undiagnosed ATTR-CM.⁴³ This multisite, open-label, single-arm trial identified 1,471 patients at risk for existing ATTR-CM by modeling predictions and included 256 of these patients by inclusion/exclusion criteria; it was able to screen 50 participants by imaging and laboratory testing. In this selected cohort, 24 of 50 patients (48%) were diagnosed with ATTR-CM. This study demonstrated the power and scalability of AI-augmented models and the very high positive screening rate in a carefully identified cohort—but also the inherent challenge of recruiting and screening patients across clinics throughout health care systems (ie, 206 of the 256 patients were excluded due to patients declining, physician not responding or declining, etc).

AREAS OF UNCERTAINTY REGARDING IMPLEMENTATION OF SCREENING STRATEGIES. Multiple questions remain related to screening; these include, but are not limited, to whether screening should be broad or focused on high-risk populations and how risk scores and prediction will be incorporated into the routine workflow of busy clinicians. Patient perception and response to identification as a “high risk” patient or a “positive” disease screening without clinical cardiac disease (ie, patient anxiety in setting of carpal tunnel screening programs) is another important aspect of population-based screening that will need to be addressed.

Though appropriate in the intent to potentially alter disease trajectory, there is limited prospective evidence to guide decision-making in the context of very early-stage disease that has been excluded from seminal trials. The management of patients with no cardiac symptoms, minimal or no elevation in N-terminal pro-B-type natriuretic peptide levels, but positive scintigraphy, for example, remains an area of debate among cardiac amyloid experts.

In a recent study, patients >50 years of age undergoing carpal tunnel decompression were recruited for addition of tenosynovial biopsy to look for

presence of amyloid protein.⁷² Of these patients, 116 biopsy-positive patients underwent noninvasive evaluation, and 32 patients had Perugini grade 1 or 2 uptake by scintigraphy. Review of the echocardiograms showed mild thickening, and biomarkers were mildly elevated in those with Perugini grade 2 (median: 367 pg/nL), prompting offering of treatment with stabilizer therapy. Those with Perugini grade 1 scan remained under annual surveillance.

Close attention to prospective follow-up data from such screening cohorts and ongoing discussions between payors, patients, and clinicians, along with considerations around cost, tolerability, and expected longitudinal benefit are warranted. Additionally, for gene-positive, asymptomatic variant patients, there is an actively enrolling trial (ACT-EARLY [Acoramidis Transthyretin Amyloidosis Prevention Trial in the Young]) that will examine the role of stabilizer therapy in asymptomatic patients.

Ultimately, to advance the work of ATTR-CM screening, there is a need for increased data on which approaches can improve clinical diagnosis in a cost-effective manner and can be incorporated practically into routine clinical practice. This raises the potential utility for future pragmatic, randomized hybrid “effectiveness-implementation” trials across multiple centers to evaluate both the performance of models and their implementation strategies.⁷¹

CONCLUSIONS

ATTR-CM remains underdiagnosed compared with population estimates, and it is increasingly recognized as a condition for which early identification and diagnosis are critical. Multiple studies have identified features that correlate with high risk of ATTR-CM and produce associated risk scores, as summarized in this

review. Despite increasing and ongoing work in this space, there remain significant gaps in quantifying individual-level ATTR-CM risk and implementing evidence-based screening strategies. Future work should better synthesize and compare the multiple existing risk models, evaluate the implementation of known risk models (with a focus on the potential downstream effects on key stakeholders including patients and clinicians), assess cost-effectiveness, and/or evaluate multifaceted approaches to leverage the described evidence base for risk features toward effective and practical health system screening strategies to ensure timely diagnosis and treatment for patients with ATTR-CM.

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